WORKED SOLUTION

STEP III 2008, Question 2

Pure • Sums of Powers

STEP III 2008, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q2

The key idea

The sum of $(r+1)^k - r^k$ telescopes, so it collapses to a single term. But we can also expand each bracket by the binomial theorem, and then the same sum becomes a combination of the lower power sums $S_j(n)$. Equating the two gives a recurrence that builds each S_{k-1} from the ones before it, and that recurrence is also exactly what we need to see the polynomial structure in part (ii).

Solution

Part (i)

First, the telescoping. Writing the sum out,

$$\sum_{r=0}^{n}\left[(r+1)^k - r^k\right] = (1^k - 0^k) + (2^k - 1^k) + \cdots + ((n+1)^k - n^k) = (n+1)^k,$$

since every intermediate term cancels and $0^k = 0$ (as $k \geq 1$).

Second, the binomial theorem. For each r,

$$(r+1)^k - r^k = \sum_{j=0}^{k-1}\binom{k}{j}r^j = 1 + \binom{k}{1}r + \binom{k}{2}r^2 + \cdots + \binom{k}{k-1}r^{k-1},$$

because the r^k terms cancel. Summing over $r = 0, 1, \ldots, n$, the constant 1 contributes $n+1$ (there are $n+1$ values of r), and each r^j with $j \geq 1$ contributes $S_j(n)$. So

$$(n+1)^k = (n+1) + \binom{k}{1}S_1(n) + \binom{k}{2}S_2(n) + \cdots + \binom{k}{k-2}S_{k-2}(n) + \binom{k}{k-1}S_{k-1}(n).$$

The last coefficient is $\binom{k}{k-1} = k$. For the others we use the symmetry $\binom{k}{j} = \binom{k}{k-j}$, so that the coefficient of $S_{k-j}(n)$ is $\binom{k}{k-j} = \binom{k}{j}$; in particular the coefficient of $S_1(n)$ is

$\binom{k}{1} = \binom{k}{k-1}$. Keeping $kS_{k-1}(n)$ on one side and moving everything else across gives

$$kS_{k-1}(n) = (n+1)^k - (n+1) - \binom{k}{2}S_{k-2}(n) - \binom{k}{3}S_{k-3}(n) - \cdots - \binom{k}{k-1}S_1(n),$$

which is $(*)$, as required. (We need $k \geq 2$ here so that S_{k-1} is one of our sums.)

Ask yourself Why did we treat the $j = 0$ term separately instead of writing it as $S_0(n)$?

S_k is only defined for positive k, and "0^0" at $r = 0$ is ambiguous. Pulling the constant out and counting it directly as $n+1$ avoids both problems, and it explains the lone $-(n+1)$ in $(*)$.

$S_3(n)$. Take $k = 4$ in $(*)$, so that $\binom{4}{2} = 6$ and $\binom{4}{3} = 4$:

$$\begin{aligned} 4S_3(n) &= (n+1)^4 - (n+1) - 6S_2(n) - 4S_1(n) \\ &= (n+1)^4 - (n+1) - n(n+1)(2n+1) - 2n(n+1). \end{aligned}$$

Every term has a factor $n+1$, so

$$\begin{aligned} 4S_3(n) &= (n+1)\left[(n+1)^3 - 1 - n(2n+1) - 2n\right] \\ &= (n+1)\left[n^3 + 3n^2 + 3n - 2n^2 - n - 2n\right] \\ &= (n+1)(n^3 + n^2) = n^2(n+1)^2. \end{aligned}$$

Hence

$$S_3(n) = \tfrac{1}{4}n^2(n+1)^2 = \big(S_1(n)\big)^2.$$

$S_4(n)$. Take $k = 5$, so that $\binom{5}{2} = \binom{5}{3} = 10$ and $\binom{5}{4} = 5$:

$$5S_4(n) = (n+1)^5 - (n+1) - 10S_3(n) - 10S_2(n) - 5S_1(n).$$

Again we take out the factor $n+1$, using our expressions for S_3, S_2 and S_1:

$$\begin{aligned} 5S_4(n) &= (n+1)\left[(n+1)^4 - 1 - \tfrac{5}{2}n^2(n+1) - \tfrac{5}{3}n(2n+1) - \tfrac{5}{2}n\right] \\ &= (n+1)\Big[n^4 + 4n^3 + 6n^2 + 4n - \tfrac{5}{2}n^3 - \tfrac{5}{2}n^2 \\ &\qquad - \tfrac{10}{3}n^2 - \tfrac{5}{3}n - \tfrac{5}{2}n\Big] \\ &= (n+1)\left[n^4 + \tfrac{3}{2}n^3 + \tfrac{1}{6}n^2 - \tfrac{1}{6}n\right] \\ &= \tfrac{1}{6}n(n+1)\left(6n^3 + 9n^2 + n - 1\right). \end{aligned}$$

Since S_4 ought to contain the factor $2n+1$ like S_2 does, we try $n = -\frac{1}{2}$ in the cubic: $-\frac{3}{4} + \frac{9}{4} - \frac{1}{2} - 1 = 0$. So $2n+1$ is a factor, and $6n^3 + 9n^2 + n - 1 = (2n+1)(3n^2 + 3n - 1)$. Therefore

$$S_4(n) = \tfrac{1}{30}\,n(n+1)(2n+1)(3n^2 + 3n - 1).$$

Part (ii)

We use strong induction on k. The claim is true for $k = 1$, since $S_1(n) = \frac{1}{2}n^2 + \frac{1}{2}n$ has degree 2 (and also for $k = 2$).

Suppose that, for some $k \geq 2$, each of $S_1(n), \ldots, S_{k-1}(n)$ is a polynomial in n with S_j of degree $j+1$. Apply $(*)$ with $k+1$ in place of k:

$$(k+1)\,S_k(n) = (n+1)^{k+1} - (n+1) - \binom{k+1}{2} S_{k-1}(n) - \cdots - \binom{k+1}{k} S_1(n).$$

On the right:

- $(n+1)^{k+1}$ is a polynomial of degree $k+1$ with leading coefficient 1;
- $n+1$ has degree 1;
- each $S_j(n)$ with $1 \leq j \leq k-1$ is, by hypothesis, a polynomial of degree $j+1 \leq k$.

So the right-hand side is a polynomial whose n^{k+1} term comes only from $(n+1)^{k+1}$ and cannot be cancelled. Dividing by $k+1 \neq 0$, $S_k(n)$ is a polynomial of degree exactly $k+1$ (with leading coefficient $\frac{1}{k+1}$). This completes the induction.

Ask yourself What does "the constant term" and "the sum of the coefficients" of a polynomial $P(n)$ correspond to?

They are $P(0)$ and $P(1)$. Write $S_k(n) = P(n) = a_{k+1}n^{k+1} + \cdots + a_1 n + a_0$; this agrees with the sum $\sum_{r=0}^{n} r^k$ for every whole number $n \geq 0$, so we may evaluate at $n = 0$ and $n = 1$.

- Constant term: $a_0 = P(0) = S_k(0) = 0^k = 0$, since $k \geq 1$.
- Sum of coefficients: $a_{k+1} + \cdots + a_1 + a_0 = P(1) = S_k(1) = 0^k + 1^k = 1$.

Both results are shown.

Checks

- Small values: $S_3(3) = 1 + 8 + 27 = 36 = \frac{1}{4} \cdot 9 \cdot 16$, and $S_4(3) = 1 + 16 + 81 = 98 = \frac{1}{30} \cdot 3 \cdot 4 \cdot 7 \cdot 35$.
- Part (ii) on our answers: $S_4(n) = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n$ has constant term 0 and coefficients summing to $\frac{6+15+10-1}{30} = 1$; its leading coefficient is $\frac{1}{5} = \frac{1}{k+1}$, as the induction predicts.
- We checked $(*)$ exactly for $2 \leq k \leq 9$ and $0 \leq n \leq 30$, and both closed forms for $0 \leq n < 200$, by direct summation.

Common slips

- Forgetting the $r = 0$ term of $\sum 1$: it contributes $n + 1$, not n.
- Expanding everything instead of taking out the common factor $n + 1$ first, which makes the algebra for S_4 much heavier.
- In part (ii), saying only that the degree is "at most $k + 1$". The question wants exactly $k + 1$, so note that the n^{k+1} term cannot cancel.

Beyond STEP

The coefficients of these polynomials are governed by the Bernoulli numbers, and Faulhaber's formula writes $S_k(n)$ as a combination of powers of n whose coefficients are binomial coefficients times Bernoulli numbers. The fact that $S_k(n)$ is a polynomial of degree $k + 1$ with leading term $\frac{n^{k+1}}{k+1}$ is the discrete analogue of $\int_0^x t^k \, \mathrm{d}t = \frac{x^{k+1}}{k+1}$, an idea made precise in the calculus of finite differences. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.